Case Report

A Rare Case Report of Bilateral Complex Macrocystic Adrenal Hemorrhage Mimicking Fetal Neuroblastoma

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Abstract

Fetal and neonatal adrenal glands are large vascular organs, which make them vulnerable to frequent bleeding. Although neonatal adrenal hemorrhage is commonly reported, it is rarely diagnosed on antenatal sonography. We present a rare case of prenatally diagnosed bilateral adrenal hemorrhage, which mimicked antenatal neuroblastoma.

Keywords: *Bilateral adrenal hemorrhage, complex cyst adrenal, fetal adrenal, neuroblastoma*

Introduction

Fetal and neonatal adrenal glands are relatively large and very well-vascularized organs in early life, which predispose them for frequent bleeding. Risk factors associated with adrenal hemorrhage *in utero* include maternal hypertension, maternal diabetes, Beckwith–Wiedemann syndrome, asphyxia, shock, infections, thrombosis of inferior vena cava and left renal vein, and hemorrhagic disorders.[1,2] Although neonatal hemorrhage is common, accounting for 1.7–2.1 per thousand births, the incidence of antenatal hemorrhage is not elicited in any study and is rarely picked up in antenatal sonography.[3] Sonographic appearance can range from simply bulky echogenic adrenals to completely cystic lesion. Sometimes, hemorrhage can complicate its appearance and may mimic neuroblastoma. Later, they can present with bilateral small adrenal glands with calcifications.

We present a rare case of prenatally diagnosed case of bilateral adrenal hemorrhage which mimicked bilateral neuroblastoma.

Case Report

A 26-year-old female gravida 1, para 0 was referred to our Imaging Centre from Obstetric Department at 32 weeks of gestation with the previous sonography report, suggesting bilateral suprarenal heteroechoic mass lesion with cystic areas, which were considered as bilateral neuroblastoma. The patient was severely hypertensive with pedal and vulval edema. Previous medical and obstetric history was unremarkable. She had no family history of congenital malformation. Detailed fetal ultrasound revealed multiloculated heterogeneous lesions with cystic areas, located superior to both kidneys (left-sided lesion larger than right). Few of the cysts showed internal debris and echogenic components. No internal vascularity was observed on the color Doppler study. Bilateral adrenal glands were not seen distinctly from the lesion. Both kidneys were displaced inferiorly and other retroperitoneal structures anteriorly. Bilateral renal veins and inferior vena cava were normal. Fetal biometry and Doppler parameters of umbilical cord were normal. There was associated polyhydramnios (AFI 19.5).

Further, antenatal magnetic resonance imaging (MRI) was acquired using T2W and HASTE sequences, which revealed complex cystic lesions in bilateral suprarenal location. Few cysts show fluid levels, suggestive of hemorrhage [Figure 1].

The patient was delivered at 32 weeks due to pregnancy-induced hypertension with a healthy male neonate weighing 1.8 kg. No anemia or palpable abdominal mass was found in the neonate. Postnatal

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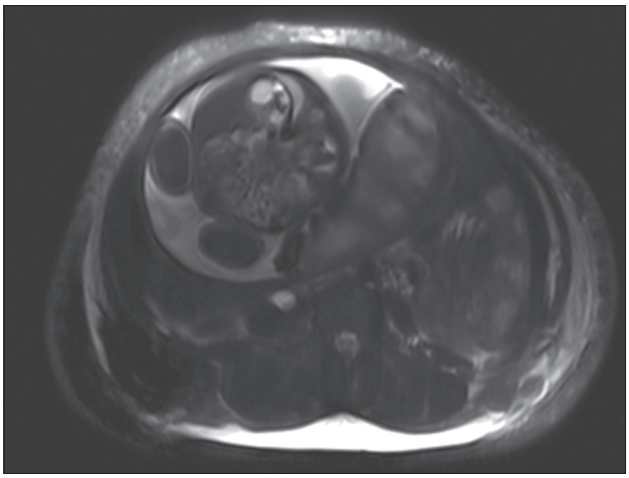


Figure 1: Antenatal magnetic resonance imaging scan: T2 weighted axial images showing heterogenous lesions in the bilateral suprarenal location with associated cystic areas

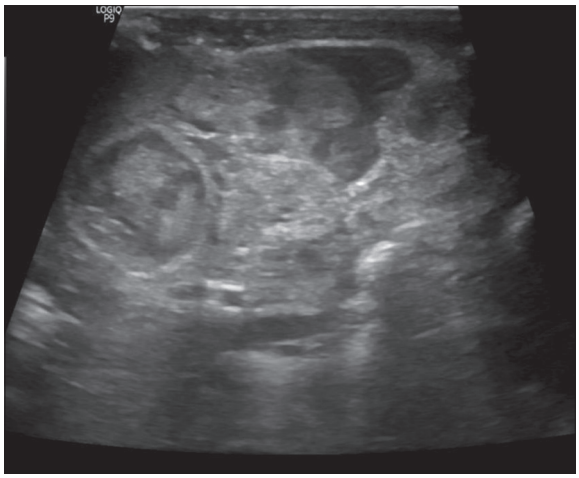


Figure 2: Postnatal sonography revealed heteroechoic lesions in the bilateral suprarenal location. Left-sided lesions showing evidence of echogenic clot and fluid levels suggestive of hemorrhage

sonography revealed heterogeneous multiloculated complex cystic lesions (left > right) [Figure 2]. All fetal hormonal evaluations including vanillylmandelic acid (VMA) urine level (0.57 μ/ml), cortisol (5.61 μ/dl), and adrenocorticotropic hormone (19.2 μ/dl) levels were within normal limits. On follow-up ultrasound scans, there was decrease in the size of bilateral lesions with appearance of cystic changes in echogenic components. Hence, finally, the diagnosis of adrenal hemorrhage was confirmed.

Discussion

Adrenal gland is supplied by three arterial branches from phrenic artery, renal artery, and directly from the aorta and so are very vascular and vital organs for human survival.[3] The main differential diagnosis of suprarenal masses in the fetus and neonate is congenital neuroblastoma, lung sequestrations, adrenal hemorrhage, mesoblastic nephroma, and duplications of the urinary or intestinal tract.[1] However, neuroblastoma is the most important differential requiring immediate attention due to its notorious extension, need for urgent therapy, and prognostic implications.

Adrenal hemorrhage is generally diagnosed by sonography during the neonatal period but is rarely detected antenatally. The prenatal diagnosis has not been definitely described, and its sonographic appearance can easily be confused with other vivid pathologic conditions. Bilaterality in adrenal hemorrhage is seen in 10%–15% cases. In unilateral lesions, right side is involved in 75% cases due to short adrenal vein on the right side.[3]

Abdominal sonography has greatly facilitated the diagnosis of adrenal hemorrhage particularly when the clinical presentation is subtle. It is the preferred screening modality because of no radiation hazard and ease of follow-ups once diagnosed. Ultrasound appearance depends on the stage of hemorrhage ranging from echogenic lesions in hyperacute stage, which liquefies shortly leading to a cystic or multicystic appearance in subacute to chronic stages of hemorrhage. Calcification can also occur as late sequelae.[4,5]

Computed tomography scan and MRI both can easily diagnose and stage adrenal hemorrhage; however, these modalities usually do not add to existing information as compared with ultrasonography.[6]

A solid or complex adrenal mass which subsequently becomes anechoic and shrinks on follow-up sonograms, prenatally and postnatally, is most consistent with adrenal hemorrhage. If this does not happen, neuroblastoma should be considered a primary differential.[5]

Neuroblastoma in the neonate and fetus can spontaneously resolve or exist *in situ*. Therefore, it is possible that adrenal calcifications, which were presumed to be sequelae of prenatal adrenal hemorrhage, can actually represent the outcome of spontaneously resolved neuroblastoma or neuroblastoma *in situ*. If visualized mass grows, fails to resolve, has internal vascularity, or has other characteristics doubtful for neuroblastoma, even with normal urinary homovanillic acid (HVA) and VMA levels, then added imaging and desirable surgical intervention should be considered. It is known that 10% of patients with neuroblastoma have normal HVA, and 27.5% have normal VMA.

We present a rare case report of prenatal adrenal hemorrhage initially detected by the ultrasound as complex cystic lesion in the bilateral suprarenal areas, and careful follow-up concluded it to be adrenal

hemorrhage. Thus, stressing the point "Not every adrenal lesion is neuroblastoma in antenatal scan," and therefore, a thorough clinical workup and follow-up can prevent undue interventions.

Declaration of patient consent

The authors certify that they have obtained all appropriate patient consent forms. In the form, the patient has given her consent for her images and other clinical information to be reported in the journal. The patient understand that name and initials will not be published and due efforts will be made to conceal identity, but anonymity cannot be guaranteed.

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Conflicts of interest

There are no conflicts of interest.

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